

Valencia Flood Mitigation Plan 2024

Introduction and Risk Assessment

Generalitat Valenciana - Conselleria d'Emergencia Climatica

1. Overview of Flood Risk in Comunitat Valenciana

The Comunitat Valenciana is one of the most flood-prone regions in the western Mediterranean. Its unique geography - steep mountain catchments draining into narrow coastal plains through dry river beds (ramblas and barrancos) - creates extreme flash flood potential. The region experiences periodic DANA events (Depresion Aislada en Niveles Altos, also known as cold drops or gota fria) that can deliver 200-400mm of rainfall in 24 hours, overwhelming drainage infrastructure.

The 34 comarcas of Valencia, Alicante, and Castellon provinces contain approximately 2.8 million buildings. Analysis under the PATRICOVA framework (Plan de Accion Territorial de caracter sectorial sobre prevencion del Riesgo de Inundacion en la Comunitat Valenciana) identifies that approximately 18% of these buildings are located in high-risk flood zones (Zone A1 or A2).

Key flood-prone areas include La Ribera Baixa (Xuquer river basin), La Vega Baja (Segura river basin), L'Horta Sud (Turia overflow), and coastal comarcas exposed to Mediterranean storm surge.

2. The DANA Phenomenon

DANA (Depresion Aislada en Niveles Altos) is a meteorological phenomenon where a pocket of cold air at high altitude becomes isolated from the jet stream and positioned over the warm Mediterranean Sea. This creates extreme atmospheric instability, producing torrential rainfall concentrated over short periods and small geographic areas.

Key characteristics of DANA events affecting Valencia:

- Extreme precipitation: 100-500mm in 6-24 hours
- Rapid onset: minimal warning time for flash floods
- Seasonal pattern: September-November peak (warm sea surface temperatures)
- Geographic concentration: coastal comarcas and river basin outlets
- Compound hazards: combined fluvial flooding, pluvial flooding, and storm surge

Climate change projections indicate DANA events may become more intense though not necessarily more frequent. Sea surface temperature increases of 1-2C amplify available moisture, potentially increasing peak rainfall intensities by 15-25% by 2050.

3. Historical Flood Events

October 2024 DANA: The most devastating flood event in modern Spanish history struck southern Valencia province on October 29, 2024. The Poyo ravine (barranco del Poyo) and tributaries of the Magro and Xuquer rivers experienced unprecedented flow rates. Affected comarcas included L'Horta Sud, La Ribera Baixa, and La Ribera Alta. Over 220 fatalities were recorded, with estimated damages exceeding 10 billion euros. The towns of Paiporta, Catarroja, Sedavi, Massanassa, and Alfafar were severely impacted.

October 1957 Turia Flood: The river Turia overflowed in Valencia city, causing 81 deaths and massive urban destruction. This event led to the Plan Sur - diverting the Turia through a new southern channel (Nuevo Cauce del Turia) completed in 1973.

October 1982 Pantanada de Tous: The Tous dam on the Xuquer river failed catastrophically, releasing 250 million cubic

meters of water into La Ribera Baixa. 30 deaths occurred and entire towns were submerged under 3-5 meters of water.

September 2019 DANA (Vega Baja): Intense DANA event affected La Vega Baja comarca in Alicante province. The Segura river overflowed, flooding Orihuela, Dolores, and surrounding municipalities. 6 fatalities and 1.5 billion euros in damages.

4. PATRICOVA Flood Zone Classification

The PATRICOVA system classifies flood risk into the following zones:

Zone A1 - Frequent Fluvial Flooding (return period < 25 years):

Areas subject to frequent flooding from rivers, ramblas, and barrancos. Mandatory restrictions on new development. Existing buildings require flood-proofing measures.

Zone A2 - Occasional Flooding (return period 25-100 years):

Areas subject to periodic flooding with moderate frequency. New development requires flood risk assessment and mitigation measures. Building standards require elevated ground floors.

Zone B - Geomorphic Flood Risk:

Areas identified through geomorphological analysis as historically subject to flooding, even if not captured by hydraulic modeling. Development permitted with conditions.

Zone C - Low Risk:

Areas with minimal flood exposure (return period > 500 years). Standard building regulations apply.

As of 2024, approximately 4,200 km² of the Comunitat Valenciana falls within Zone A1 or A2, affecting 34 comarcas to varying degrees.

5. Current Flood Protection Infrastructure

Major flood defense infrastructure in the Comunitat Valenciana includes:

- Nuevo Cauce del Turia: 12km engineered channel diverting the Turia river south of Valencia city. Capacity: 5,000 m³/s. Completed 1973.
- Tous Dam (rebuilt): Reconstructed after 1982 failure with improved spillway capacity. Reservoir volume: 80 hm³.
- Segura River channelization: Concrete-lined channel through Orihuela and Vega Baja. Design capacity: 400 m³/s (exceeded during 2019 DANA).
- Albufera drainage network: Pump stations and canal system managing water levels in L'Horta Sud and La Ribera Baixa.
- SAIH Jucar: Real-time hydrological monitoring network with 145 rainfall stations and 82 river flow gauges across the Xuquer basin.
- AEMET early warning system: National meteorological agency issuing color-coded alerts (yellow/orange/red) for DANA and heavy rainfall events.

Despite this infrastructure, the October 2024 event demonstrated that existing defenses are insufficient for extreme DANA scenarios, particularly in the barranco del Poyo catchment which had no major flood defense works.